

DOT/FIL OU Pairs Trading

Hai Yen Vu

This post is a short case study on pairs trading using hourly crypto perpetual data. The question is simple:

If the DOT/FIL spread looks mean-reverting, can an OU score give a cleaner trading signal than a rolling z-score?

The tested pair is PERP:DOT/USDT and PERP:FIL/USDT. I use the same spread definition for both methods. The rolling z-score baseline measures the spread against a local rolling mean, while the OU score measures it against an equilibrium level estimated from an Ornstein-Uhlenbeck process.

The main result is that both methods are positive on the final test set, but the OU score has better return, lower drawdown, and better profit factor in this sample. The result is still not strong enough because the test Sharpe is below 1.

1 1. Research split

The notebook uses a chronological split:

```
train_frac = 0.70
formation_frac = 0.40
```

Segment	Period	Purpose
Formation	2022-01-01 -> 2023-03-28 01:00	Fit hedge ratio and initial spread model
Validation	2023-03-28 02:00 -> 2025-02-01 16:00	Select threshold/window
Test	2025-02-01 17:00 -> 2026-05-31	Final out-of-sample check

2 2. Building the DOT/FIL spread

The spread is built from a log-price regression:

$$Z_t = \log P_t^{DOT} - \hat{\alpha} - \hat{\beta} \log P_t^{FIL}$$

The notebook code is direct:

```
def fit_price_spread(log_price_frame, left_symbol, right_symbol):
    pair_log = log_price_frame[[left_symbol, right_symbol]].dropna()
    y, x = pair_log[left_symbol], pair_log[right_symbol]

    xmat = np.column_stack([np.ones(len(pair_log)), x])
    alpha, beta = np.linalg.lstsq(xmat, y, rcond=None)[0]
    spread = y - alpha - beta * x

    return float(alpha), float(beta), spread
```

In the formation window:

Metric	Value
alpha	0.542228
beta	0.794317
cointegration p-value	0.001850
ADF p-value	0.000310
AR(1) half-life	161.61 bars

The cointegration and ADF diagnostics are both strong in formation, and the half-life is long but still usable for an hourly mean-reversion case.

3 3. Rolling z-score vs OU score

The rolling z-score is:

$$z_t = \frac{Z_t - \bar{Z}_{t,L}}{s_{t,L}}$$

It is simple, but the signal depends heavily on the rolling window (L). If the window is too short, the score can overreact. If it is too long, the signal can become slow.

The OU model assumes:

$$dZ_t = -A(Z_t - m)dt + v dW_t$$

After fitting the OU process, the score is:

$$OU_score_t = \frac{Z_t - m}{v/\sqrt{2A}}$$

This asks a different question: how far is the spread from its estimated long-run equilibrium in stationary OU standard deviations?

4 4. Validation calibration

The notebook does not hard-code the final threshold. It tests a small grid:

```
entry_grid = [1.25, 1.50, 1.75, 2.00, 2.25, 2.50]
stop_mult_grid = [1.5, 2.0, 2.5, 3.0]
rolling_window_grid = [72, 120, 168, 336]
```

Then it selects the best validation config separately for each method:

The selected validation parameters are:

Strategy	Entry	Window	Stop	Validation return	Validation Sharpe	Entries
Rolling z-score	2.50	168	6.25	46.62%	1.05	93
OU score	1.50	n/a	4.50	99.03%	1.78	26

5 5. Trading rule

The entry and exit logic is the same for both scores:

```

if position == 0:
    if score <= -entry:
        position = 1      # long spread
    elif score >= entry:
        position = -1     # short spread
else:
    if mean_reverted or stopped or timeout:
        position = 0

```

The position is mapped into two hedge-normalized legs:

```

weights = pd.DataFrame({
    symbol_a: position / (1 + abs(beta)),
    symbol_b: -position * beta / (1 + abs(beta)),
})

```

6 6. Test result

On the final test segment:

Strategy	Entry	Window	Stop	Test return	Sharpe	Max DD	Position cover- age	Total trades	Profit factor	Fees paid
Rolling z- score base- line	2.50	168	6.25	13.53%	0.55	26.61%	40.79%	2607	1.08	0.0580
OU score	1.50	n/a	4.50	24.60%	0.85	14.82%	59.62%	3070	1.21	0.0290

The OU fit before the test gives:

OU metric	Value
OU half-life estimated before test	252.07 bars
stationary standard deviation	0.088461

In this test, it improves return and drawdown even though it is not less exposed

7 7. Conclusion

There are four reasons to stay careful:

1. This is still one pair and one main split.
2. The split and pair were found during research, so there is some research selection bias.
3. The test Sharpe is below 1, so the result is profitable but not statistically comfortable.
4. Funding, order-book liquidity, and stronger slippage modeling are not included.

So my conclusion for this is in this DOT/FIL case study, validation-calibrated OU scoring gives a better OOS result than a validation-calibrated rolling z-score baseline. The next step should be walk-forward testing across multiple candidate pairs selected by the same rule.